

Shivraj Dave (He/Him)

Windsor, ON (Ready to Relocate) | 437-696-1901 | shivrajdave99@gmail.com | linkedin.com/in/shivrajdave

Professional Summary

Electrical Engineer with Master of Engineering (ECE) and 3+ years of progressive experience in hardware validation, embedded systems testing, and manufacturing operations. Core expertise spans battery management systems, Hardware-in-the-Loop (HIL) testing, CAN bus communication protocols, and regulatory compliance documentation (FDA/USFDA). Proven ability to bridge complex control system design with hands-on production troubleshooting. Skilled in Python, MATLAB/Simulink, PLC programming (Delta PLCs), and precision instrumentation testing in regulated environments. Seeking to leverage battery systems expertise, test automation fundamentals, and industrial validation discipline in automotive manufacturing and controls engineering roles.

Technical Skills & Proficiencies

- **Control Systems & Embedded Testing:** PLC Programming (Ladder Logic, Delta PLCs) | Hardware-in-the-Loop (HIL) Testing | ECU Integration | Battery Management Systems (BMS) | Control System Co-Simulation | Embedded Systems Troubleshooting
- **Automotive & Power Systems:** Battery Pack Design & Architecture | Electric Vehicle (EV) Powertrains | CAN Bus Communication Protocols | High-Voltage Systems | Propulsion System Integration
- **Software & Scripting:** Python (PyTorch, Data Processing, Algorithm Implementation) | MATLAB/Simulink | AutoCAD Electrical | Revit
- **Test & Validation:** Instrument Commissioning & Calibration | Requirements Traceability | Test Procedure Development | Data Acquisition & Analysis | Root Cause Analysis
- **Quality & Regulatory:** FDA/USFDA Documentation Standards | ISO13320 | USP <429> | ICH Q2(R2) | 21 CFR Part 11 | IEC60825-1:2007 | Regulatory Compliance Frameworks

Professional Experience

University of Windsor | Windsor, ON

January 2025 – April 2026

MEng Research Project: Adaptive Battery Management System (BMS) Control for Mismatched HEV Modules

- Designed and implemented adaptive control algorithms for hybrid electric vehicle battery pack management using co-simulation approach
- Developed MATLAB/Simulink models integrating real-time battery dynamics with controller logic to manage mismatched module performance
- Established requirements traceability documentation linking system specifications through control logic validation and performance testing
- Created Python scripts for data processing, model validation, and performance metric analysis
- Produced technical documentation validating control architecture against system requirements

Charotar University of Science and Technology | India

February 2020 – May 2021

Bachelor's Capstone Project: Electric Vehicle (EV) Research & Development

- Designed battery pack architecture with thermal and electrical integration for EV platform
- Developed Hardware-in-the-Loop (HIL) test framework for ECU validation against vehicle dynamics models
- Configured CAN bus communication between battery management, motor controller, and onboard diagnostics
- Established test procedures and validated performance against design specifications

- Executed commissioning and validation protocols for laser diffraction particle size analyzers, X-ray analytical equipment, and spectrometry systems across 75+ customer accounts
- Performed rigorous instrument qualification testing across 150+ pharmaceutical plants, 200+ chemical facilities, and 20+ mineral processing sites
- Achieved 95%+ first-pass validation success rate through systematic troubleshooting and precision calibration
- Developed traceability documentation for all testing procedures, calibration data, and equipment performance metrics
- Ensured compliance with USP <429>, ISO13320, ICH Q2(R2), 21 CFR Part 11, and IEC60825-1:2007 standards
- Applied advanced knowledge of scattering theory to optimize equipment performance and troubleshoot system failures
- Worked extensively with CAN bus communication protocols, device networking, and embedded firmware updates
- Executed 270+ successful instrument calibrations, diagnostics, and emergency resolutions (23 site visits/month average, 85% travel)

Linamar Corporation | Windsor, ON

April 2025 – April 2026

Production Engineering Support – Automotive Manufacturing

- Operated high-precision automated manufacturing systems (Okuma Degate) in high-volume production, processing 350+ parts/shift at 1-minute cycle time
- Collaborated with production, quality, and controls teams to identify and resolve quality issues in 10R80 transmission components
- Performed precision leak testing and microscopic failure analysis on cylinder head components
- Identified misclassifications in automated leak detection systems through data-driven analysis
- Supported continuous improvement initiatives focused on downtime reduction and production optimization
- Gained hands-on exposure to industrial automation systems, PLC controls, and real-time equipment monitoring

Education

University of Windsor | MEng in Electrical and Computer Engineering

January 2025 – April 2026

Charotar University of Science and Technology | B.Tech Electrical Engineering July 2017 – June 2021

Certification and Technical Training

- Autodesk AutoCAD Electrical for Electrical Designers
- Electric Vehicles - Part 1 (NPTEL)
- Electrical Systems: Panel Boards, Frequency Drives, and Transformers
- Malvern Panalytical Laser Diffraction Particle Size Analyzer & Accessories
- Standard First Aid and CPR-C with AED
- Verified International Academic Qualifications (WES)

Professional Affiliations

- **Affiliate Member & Event Organizer** | Ontario Society of Professional Engineers (OSPE)
- **President** | Gujarati Students Association (GSA), University of Windsor
- **Community Volunteer** | Hindu Temple and Cultural Centre of Windsor